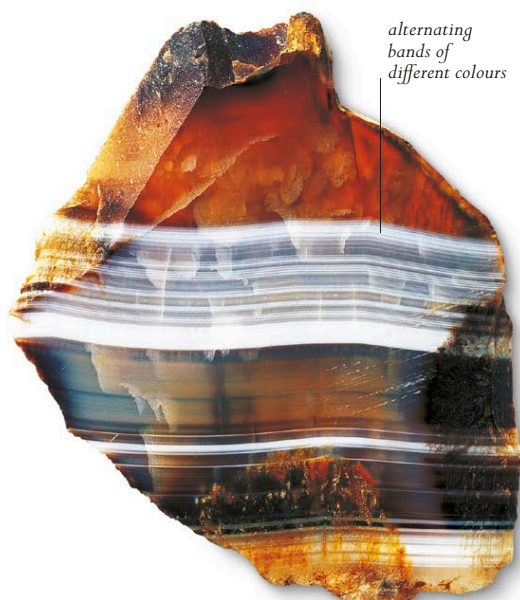


» TECTOSILICATES



*alternating
bands of
different colours*

OPAL

Amorphous

H 5½–6½ • SG 1.9–2.3

Opal is a hydrated silicon dioxide, occurring as nodules, encrustations, or masses in most rock types.

Impurities impart a variety of colours. Opal is treasured as a gemstone.

**CHALCEDONY**

Trigonal

H 7 • SG 2.65

Chalcedony is microcrystalline quartz, a silicon dioxide. Pure chalcedony is white. It forms in veins and cavities in many rock types.

ONYX

Trigonal

H 7 • SG 2.7

Onyx is a parallel-banded gemstone variety of chalcedony. It is not particularly common; notable locations are India and S. America.

**CARNELIAN**

Trigonal

H 7 • SG 2.7

Carnelian is a variety of chalcedony coloured red to orange by iron oxide. The finest quality carnelian comes from India.

**BLOODSTONE**

Trigonal • H 7 • SG 2.7

Bloodstone is a variety of chalcedony coloured dark green by traces of iron silicates. Flecks of red jasper throughout resemble blood.

**CHRYSOPRASE**

Trigonal • H 7 • SG 2.7

Chrysoprase is a variety of chalcedony containing nickel, which imparts a green colour. It is the most valuable of the chalcedony varieties.



*precious
opal*



*streaks of
yellow opal*

yellow cancrinite

prismatic crystal

**CANCRRINITE**

Hexagonal

H 5–6 • SG 2.42–2.51

The feldspathoid cancrinite is a variously coloured hydrated sodium calcium aluminosilicate carbonate sulfate.

**MICROCLINE**

Triclinic

H 6–6½ • SG 2.54–2.57

A very common alkali feldspar, this potassium aluminosilicate is usually white or pinkish. The green variety is called amazonstone.

**ANORTHITE**

Triclinic

H 6–6½ • SG 2.74–2.76

This plagioclase feldspar is a calcium aluminosilicate. It forms pale-coloured crystals, grains, or masses.

**HEULANDITE**

Monoclinic

H 3–3½ • SG 2.2

This zeolite is a hydrated sodium calcium aluminosilicate. It is used in petroleum refining as a molecular sieve.

**SCAPOLITE**

Tetragonal

H 5½–6 • SG 2.5–2.78

The name scapolite encompasses a series of complex sodium calcium silicates, occurring mainly in metamorphic rocks.

ironstone matrix

